TinyML Keyword Spotting on ARM Cortex-M

Lightweight keyword spotting (KWS) system for microcontrollers using TensorFlow Lite Micro, CMSIS-DSP, and CMSIS-NN.
Detects a single keyword ("yes") from 1-second audio windows with ultra-low latency.

Metrics & Performance Summary

Metric	Value
Model type	CNN (6 Conv layers + GAP + Softmax)
Parameters	~1,000
Input	61 frames × 20 log-mel coefficients (~1s audio)
Quantization	int8
Inference time	~60 ms on Cortex-M4F 64 MHz
Training accuracy	~90%+ (3 epochs)
Memory footprint	Very small, fits on low-power MCUs

Highlights

- Approximately 1,000 parameter CNN
- ~60 ms inference on Cortex-M4F @ 64 MHz
- int8 quantized model
- CMSIS-NN accelerated CNN
- CMSIS-DSP accelerated feature extraction
- Designed for low-power Cortex-M MCUs

Project Overview

Keyword spotting is a lightweight speech recognition system for detecting specific trigger words.

This project implements single-keyword spotting using the Google Speech Commands dataset, deployed on an ARM Cortex-M4F MCU.

Classes

- `keyword` ("yes")
- `unknown speech`
- `silence`

Effectively acts as binary classification:
`keyword vs everything else`

The keyword "YES" is chosen for its two phonemes, improving single-word detection reliability.

Model Summary

Feature	Value
Model type	Convolutional Neural Network (CNN)
Parameters	~1,000
Layers	6 convolution layers
Pooling	Global Average Pooling
Output	Softmax
Quantization	int8
Input	Log-Mel Spectrogram
Framework	TensorFlow / TensorFlow Lite Micro

Input Features

Audio Parameters

Parameter	Value
Sample Rate	16 kHz
Frame Size	512 samples
Frame Duration	32 ms
Hop Size	256 samples
Hop Duration	16 ms
Overlap	50%

Spectrogram Dimensions

61 time frames × 20 log-mel coefficients (~1 second of audio)

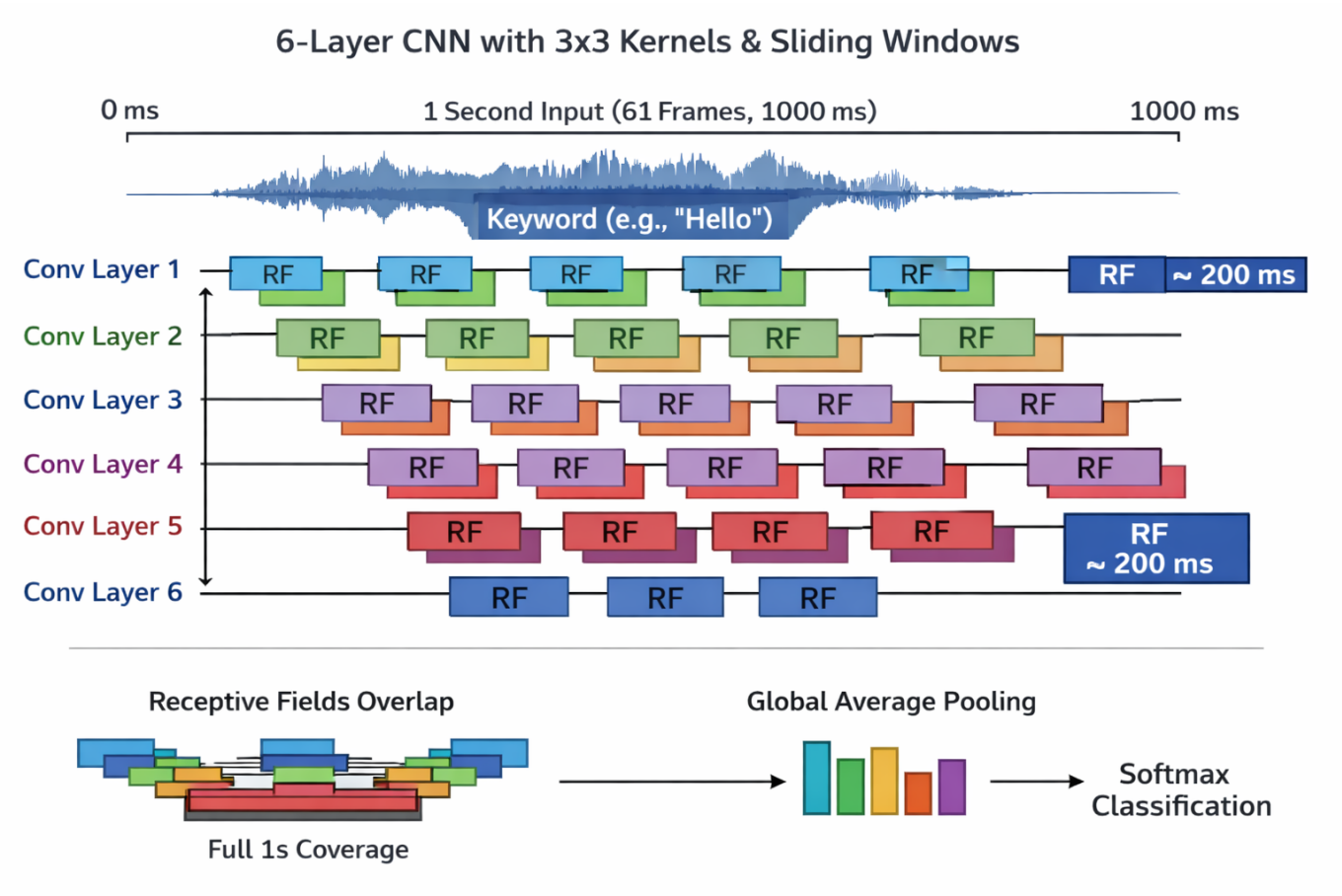
Model Architecture

- 3 convolutional stages: Conv2D → DepthwiseConv2D
- Global Average Pooling
- Softmax output

```
Input Spectrogram
    ↓
```

```
Conv2D → DepthwiseConv2D
    ↓
Conv2D → DepthwiseConv2D
    ↓
Conv2D → DepthwiseConv2D
    ↓
Global Average Pooling
    ↓
Softmax
```

Stacked layers increase the receptive field, capturing long-range phonetic patterns.



Embedded Deployment

Hardware

- ARM Cortex-M4F @ 64 MHz
- Hardware FPU for DSP acceleration

Libraries

- CMSIS-DSP: FFT, vector/matrix math
- CMSIS-NN: accelerated CNN kernels

TensorFlow Lite Micro automatically invokes CMSIS-NN for optimal performance.

Feature Extraction Pipeline

```
arm_mult_f32(audioFrame, hannWindow, windowed, FRAME_SIZE);
arm_rfft_fast_f32(&fft_inst, windowed, fftOut, 0);

for (int i = 0; i < SPECTRUM_BINS; i++) {
    float real = fftOut[2*i];
    float imag = fftOut[2*i+1];
    mag[i] = real*real + imag*imag;
}

arm_mat_mult_f32(&melMat, &magVec, &melOut);

for (int m = 0; m < NUM_MEL; m++) {
    float log_mel = logf(melRowTmp[m] + 1e-6f);
    int32_t q = (int32_t)roundf(log_mel / input_scale + input_zero_point);
    quantized_mel_row[m] = (int8_t)constrain(q, -128, 127);
}
```

Manual power spectrum ensures consistency between training and MCU pipeline.

Performance

Training

Metric	Value
Accuracy (3 epochs)	~90%+
Dataset	Google Speech Commands

Inference

MCU	Inference Time	Parameters
Cortex-M4F 64MHz	~60 ms	~1,000

Optimization could reduce inference to 20–30 ms.

Benchmark & Comparison

Model	Dataset	Parameters	Inference (ms)	Accuracy	MCU
TinyML KWS (this project)	Google SC	~1,000	60	~90%+	Cortex-M4F 64MHz

Model	Dataset	Parameters	Inference (ms)	Accuracy	MCU
TFLM CNN Example	Google SC	~8,000	120	92	Cortex-M4F 72MHz
ARM CMSIS-NN Ref	Google SC	~2,500	45	88	Cortex-M4 80MHz
KeywordSpotting-Small	Google SC	~10,000	150	95	STM32F746 216MHz

TinyML KWS is ultra-lightweight and maintains real-time operation.

Batch vs Streaming Inference

- Batch mode: ~1 s latency
- Streaming mode (future): 30–100 ms latency

Circular buffers can reduce DSP overhead and improve streaming latency.

Silence Handling

Limited dataset silence → weak correlation with real silence.

Solution: augment silence data and retrain.

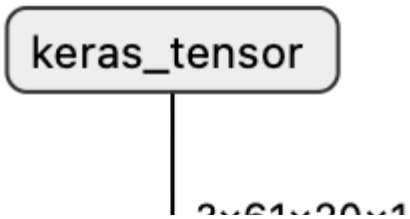
Future Work

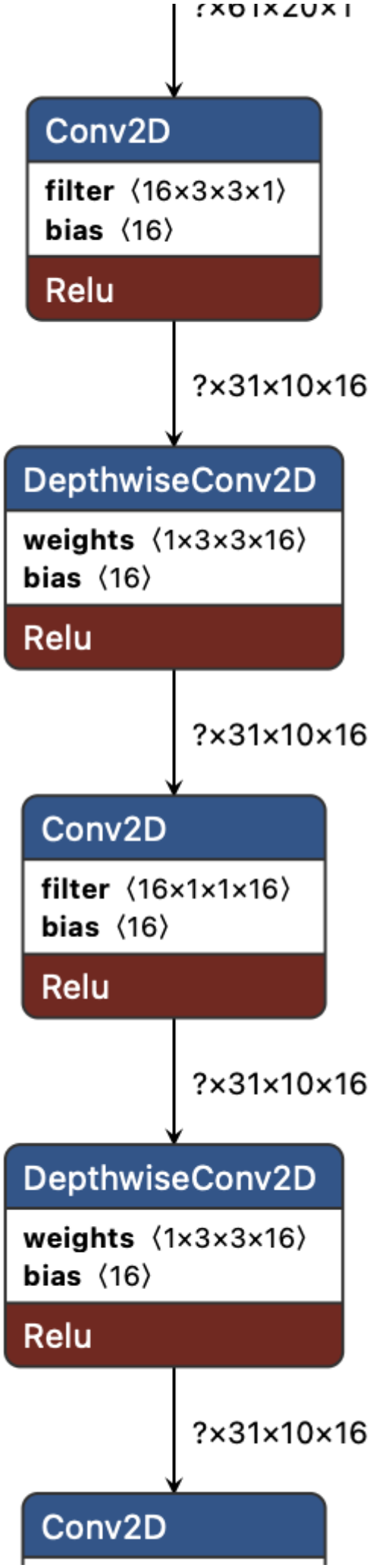
1. Streaming audio pipeline (circular buffers)
2. Multi-keyword detection (5–10 keywords)
3. TinyLLM voice interface with flexible commands:

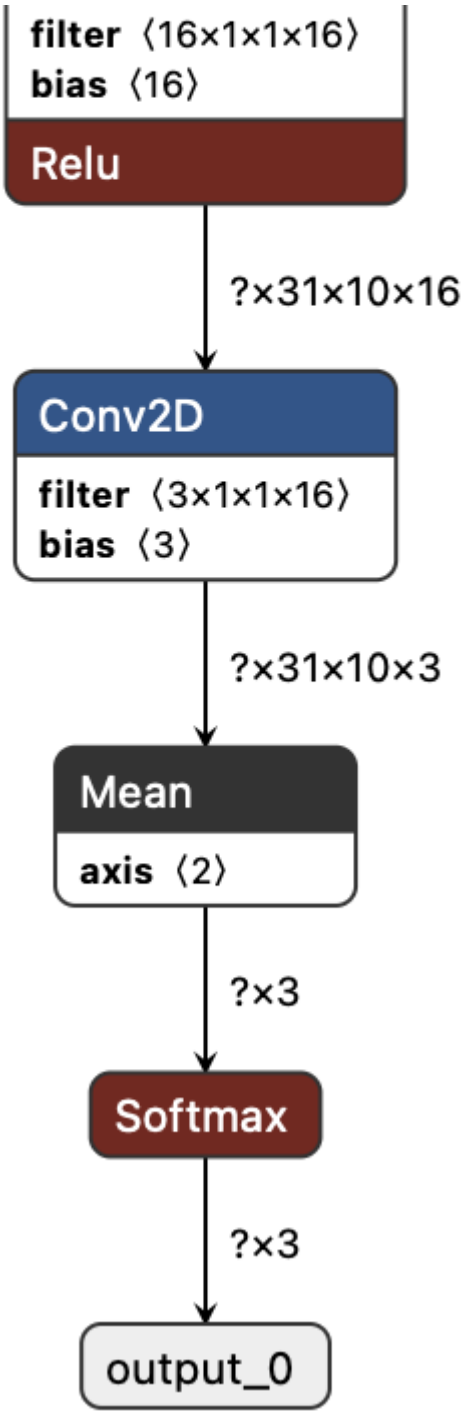
```
Robot, get me a black coffee
Robot, I want a coffee
Robot, I'll take my coffee now
```

Model Visualization

Generated using Netron:




?×61×20×1
Conv2D
filter ⟨16×3×3×1⟩
bias ⟨16⟩
Relu
?×31×10×16
DepthwiseConv2D
weights ⟨1×3×3×16⟩
bias ⟨16⟩
Relu
?×31×10×16
Conv2D
filter ⟨16×1×1×16⟩
bias ⟨16⟩
Relu
?×31×10×16
DepthwiseConv2D
weights ⟨1×3×3×16⟩
bias ⟨16⟩
Relu
?×31×10×16
Conv2D



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